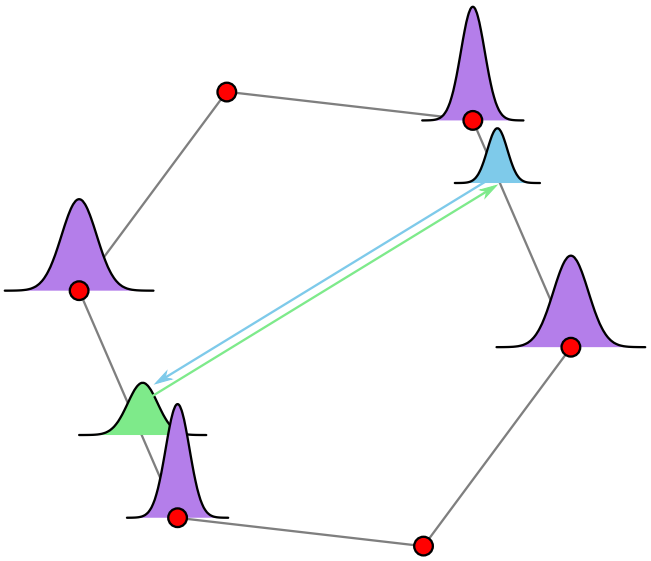


Final Report

The Art of Scientific Computing: Project Title: A selection of granary queueing models

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Subject Handbook code:
COMP-90072



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Original Version: 1st March 2018

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Chapter 1

Introduction

In which we talk about granaries and justify the importance of the question

Chapter 2

Background

In which we cover basic queueing theory, classify our problem and mention the closed form results that are going to be important for verification.

Chapter 3

Models

In which we discuss the three models of a granary we're going to be using

First the shared infrastructure of a simple queue and a single deterministic server

- An open loop with Markov modulated Poisson arrival
- A closed loop with an abstracted truck fleet and an abstracted grain supply
- A simulation of all the farms in the catchment each with their own trucks

Chapter 4

Implementation

In which we discuss the implementation in python. Discuss the initial simpy profiling, and the subsequent handmade simulation with a queue using a deque and a priority queue of upcoming events using a heapq. Discuss parallelisation with joblib - embarassingly parallel

Chapter 5

Validation

Out come the closed form results, most useful for model 1. In correct regimes model 2 and 3 should converge to something like those results as well

Chapter 6

Experiments

How do mean and tail waiting times change as we increase weather variance?
- How does overtime increase as we increase weather variance?
- How do the models compare to one another?
- Report on parallelisation efficiency

Chapter 7

Results and Discussion

When models are tuned to be similar, do the tunings make sense analytically?
Where do models disagree when tuned to be similar?

Chapter 8

Conclusion